# Likhith Gadde


Embedded Systems Engineer – Robotics & Industrial Systems

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## Professional Summary

Embedded systems engineer with experience designing and deploying sensor-driven embedded systems in electrically noisy industrial environments. Strong background in hardware-firmware integration, communication reliability, and system robustness under real-world constraints including long cable runs, unstable power, and field variability. Experienced in building end-to-end systems spanning PCB design, firmware, enclosure design, mechanical design, deployment, and root-cause debugging through measurement-driven analysis.

## Core Skills

**Embedded & Firmware:** C, C++, Python, ESP32, Raspberry Pi, FreeRTOS, inter-core sync, I2C, SPI, UART, MQTT, CAN
**Hardware Design:** KiCad (PCB), FreeCAD (CAD Design), mixed-signal 2-layer PCB design, sensor interfacing, ADS1115, MCP-series ADCs, enclosure design, 3D printing
**Signal Integrity & Debugging:** I2C bus stability, pull-up tuning, bus segmentation, oscilloscope-based validation, EMI/noise troubleshooting, field debugging
**Systems Engineering:** hardware-software co-design, sensor calibration, validation, DFM/DFA, BOM optimization, supplier evaluation, industrial deployment, familiar with CAN fundamentals (termination, differential signaling)

## Professional Experience

**Mechanical Design Engineer (Contract)** — Apr 2026 – Present
*Glarenergy* — *[Hyderabad, India]*

- Develop mechanical solutions for defined problem statements, including mechanism design, part design, and system-level integration.
- Design components and assemblies with focus on DFM, DFA, fitment, and structural reliability.
- Execute rapid prototyping workflows using 3D printing, iterating designs based on functional testing and assembly constraints.
- Build and test physical prototypes, validating performance and refining designs based on real-world behavior.

**Lead Engineer — IoT and Hardware Systems** — Dec 2024 – Mar 2026
*KLVIN Technology Labs* — *Hyderabad, India*

- Led development of industrial embedded systems from architecture through deployment across multiple factory environments.
- Designed and validated custom mixed-signal PCBs integrating sensors, GSM modules and embedded processing units.
- Analyzed and resolved communication instability in multi-node embedded systems with long cable runs (>1 meter), identifying timing degradation due to parasitic effects and implementing architectural changes to restore reliability.
- Investigated and resolved multiple field failures across deployed systems, including communication instability, sensor faults, and network connectivity edge cases.
- Reduced data transmission latency from approximately 7 seconds to 0.3 seconds per packet by restructuring system architecture: migrating from 2G to 4G, optimizing payload size, and parallelizing acquisition and transmission using dual-core FreeRTOS.
- Designed firmware architecture for concurrent data acquisition and communication, ensuring stable operation under variable sensor and network conditions.
- Improved measurement reliability by integrating external ADCs to address MCU limitations, including input impedance sensitivity and instability.
- Designed and iterated mechanical components and assemblies using CAD and in-house 3D printing, including enclosures, gaskets, grommets, and custom mounting parts; optimized for sealing, durability, and deployment constraints.
- Optimized Bill of Materials (BOM) for cost, availability, and manufacturability, achieving an estimated 31% reduction in unit cost.

**IoT & Hardware Engineer** — Jun 2024 – Nov 2024
*KLVIN Technology Labs* — *Hyderabad, India*

- Developed multi-sensor embedded systems with wireless communication for industrial monitoring applications.
- Developed and debugged PCB based systems integrating sensors, power and communication interfaces under real-world constraints.
- Performed sensor calibration and validation using measurement-driven methods (oscilloscope based analysis), deriving empirical models for accurate data interpretation.
- Identified and mitigated sensor instability and communication inconsistencies through hardware and firmware modifications.

**Engineering Intern — Computer Vision** — Nov 2023 – May 2024
*Harvested Robotics* — *Hyderabad, India*

- Developed a real-time weed detection system using YOLOv8 for autonomous agricultural robots.
- Collected field data and built a custom dataset, performing annotation using CVAT for supervised training.
- Trained object detection models on the dataset and integrated the resulting weights into the application for real-time inference.
- Contributed to system integration involving computer vision, embedded systems, and mechanical components.

## Selected Projects

**Autonomous Stabilized Targeting Turret** — Mahindra University

- Designed and built a shoulder-mounted, self-stabilizing vision-guided turret with closed-loop orientation control and autonomous target alignment.
- Implemented a distributed control architecture using Raspberry Pi for computer vision and high-level logic with a microcontroller for real-time orientation and servo control.

- Implemented IMU-based stabilization using sensor fusion (using Mahony filter) to reduce drift and maintain orientation accuracy.
- Integrated object detection, embedded electronics, and control software into a functional end-to-end prototype.

**Entry Logger using MQTT** — Mahindra University

- Built an IoT system that identified employees from RFID scans and displayed names and custom messages on an LCD.
- Used MQTT for real-time communication across connected devices.
- Integrated sensing, wireless communication, and embedded display logic into a complete prototype.

## Education

**B.Tech — Electronics and Computer Engineering** — 2020 – 2024
*Mahindra University* — *Hyderabad, India*

## Courses & Certifications

**Mechanism and Robot Kinematics** — NPTEL, IIT Kharagpur — Aug 2025 – Oct 2025
**Human Error Mitigation Workshop** — Confederation of Indian Industry — Jul 2025
**PG Level Advanced Certification Programme in Sensor Technologies** — IISc Bengaluru — Aug 2024 – Jan 2025

## Additional Experience

**Teaching Assistant — Programmable Devices** — Aug 2023 – Dec 2023
*Mahindra University* — *Hyderabad, India*

- Facilitated student learning through guided paper discussions, quiz support, and doubt-clearing sessions.

**Teaching Assistant — Design Thinking** — Jan 2023 – Jun 2023
*Mahindra University* — *Hyderabad, India*

- Supported instructors and coordinated student activities for the Design Thinking course.

**Research Intern — Smart Farming Infrastructure** — Nov 2022 – May 2023
*Mahindra University* — *Hyderabad, India*

- Designed an autonomous micro-bot for smart farming applications and presented the work at the Technology Vision 2047 student conclave at IISc Bangalore, where my team secured first place.

## Additional Information

**Tools:** Oscilloscope; familiar with logic analyzers; Git, Linux, KiCad, FreeCAD, Blender, ESP32, Raspberry Pi, NVIDIA Jetson
**Leadership & Activities:** Media Club President, Erudite Literary Club Mentor and Design Head, Aether-23 Organizing Committee
**Achievements:** First place at the Technology Vision 2047 student conclave at IISc Bangalore